The Limits of Explanation

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Abstract

Train a machine learning classifier to distinguish colon from kidney tumours using 10,935 gene-expression features, and the algorithm names TSPAN8—a gene in the invasion/metastasis pathway—as the single most important biomarker. Change nothing but the random seed and retrain: now the top gene is CEACAM5, an immune-evasion marker with zero shared Gene Ontology biological process terms. We prove that this instability is not a software bug: it is a mathematical certainty. Whenever observationally equivalent configurations coexist (the Rashomon property), no explanation can simultaneously be faithful, stable, and decisive. We derive this impossibility from first principles in four high-stakes domains—genomics, mechanistic interpretability of neural networks, causal inference, and neuroimaging—each requiring zero shared axioms. In each domain, a constructive resolution exists: projecting onto the symmetry-invariant subspace. For neural circuits, this projection lifts inter-model agreement from $\rho = 0.52$ to $\rho = 0.93$. For neuroimaging, orbit averaging across 48 analysis teams is near-optimal, with 16 independent analyses sufficing for 95% stability. That eight scientific communities—in ML, crystallography, gauge theory, causal inference, statistical mechanics, genetics, linguistics, and database theory—independently converged on this same resolution, without cross-pollination, is now explained: the orbit-averaged projection is the unique Pareto-optimal strategy for any system with the Rashomon property. The framework is verified in Lean 4 (491 theorems, 0 unproved goals) and generates quantitative predictions confirmed across seven domains ($R^2 = 0.96$, zero free parameters).

Introduction

A biologist following the first model into the wet lab would pursue invasion signalling through TSPAN8's tetraspanin-enriched microdomains; a biologist following the second would pursue immune evasion through CEACAM5's role in cell–cell adhesion. Both models predict patient labels with indistinguishable accuracy. The alternation replicates on an independent tissue contrast (endometrium vs colon: TSPAN8 78%, CEACAM5 22%) and on a different gene pair (breast vs colon: COL3A1 72%, COL1A1 28%), where both collagens share the extracellular matrix pathway—two modes of instability, one within and one between biological pathways, both governed by the same theorem.

The same structure appears wherever a system admits multiple valid explanations. Over the past century, eight scientific communities have independently converged on a single mathematical strategy for such problems: project onto the subspace invariant under the symmetry of equivalent configurations. Crystallographers who cannot recover phase from diffraction amplitudes use Patterson maps. Causal modellers who face Markov-equivalent graphs report CPDAGs. Gauge theorists quotient out local symmetry. Neuroimaging teams who reach different conclusions [Botvinik-Nezer et al., 2020] compute overlap maps. Ten neural networks trained on the same task discover circuits

that overlap by only 2.2%, yet all achieve 100% accuracy. That these communities arrived at structurally identical solutions without any cross-pollination demands an explanation.

Here we supply one. A single theorem, requiring only the Rashomon property [Breiman, 2001] as a hypothesis, proves that no explanation of an underspecified system can be simultaneously faithful (reflect the system's actual structure), stable (consistent across equivalent configurations), and decisive (commit to a specific answer). The limits of explanation are structural, not methodological—and the resolution these communities discovered is the unique optimal response. Prior work has documented this instability empirically [Krishna et al., 2022] and proved special cases for Shapley-based methods [Bilodeau et al., 2024]; here we prove the general impossibility for *any* explanation type, characterise the complete design space, and show that the same structure governs four high-stakes domains (Fig. 1). The proof is verified in Lean 4 with zero domain-specific axioms. A strengthened form—the bilemma—shows that for binary explanations (significant or not, positive or negative), even faithful and stable together are impossible. The complete design space is characterised by a tightness classification: whether the impossibility admits a resolution depends on whether the explanation space contains a neutral element (Table 1).

Results

The impossibility

An *explanation system* is a tuple $(\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\mathbf{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, obs maps configurations to observables, exp maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. The system has the *Rashomon property* if there exist $\theta_1, \theta_2 \in \mathbf{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$.

Theorem 1 (Impossibility) *No explanation system with the Rashomon property admits an explanation map that is simultaneously faithful, stable, and decisive.*

The proof is four lines (Methods). Each property is load-bearing: dropping any one yields a two-property map that is constructively achievable (Extended Data Table 1).

For binary $\mathcal{H}$ (e.g. SHAP sign, gene significance, edge orientation), the bilemma strengthens this: faithful + stable alone is impossible when every pair of distinct explanations is incompatible. This triggers for SHAP sign (positive vs negative attribution), feature selection (selected vs not), and counterfactual direction.

Tightness classification

Whether the impossibility admits a resolution depends on the structure of $\mathcal{H}$. A *neutral element* $h_0 \in \mathcal{H}$ is compatible with every other element. A *committal element* is incompatible with at least one.

Table 1: **Tightness classification.** Which property pairs are achievable depends on the existence of neutral and committal elements in $\mathcal{H}$. The bilemma occupies the bottom-right cell.

	Committal $\exists$	No committal
Neutral $\exists$	F+D, F+S, S+D	F+D, F+S
No neutral	F+D, S+D	**F+D only**

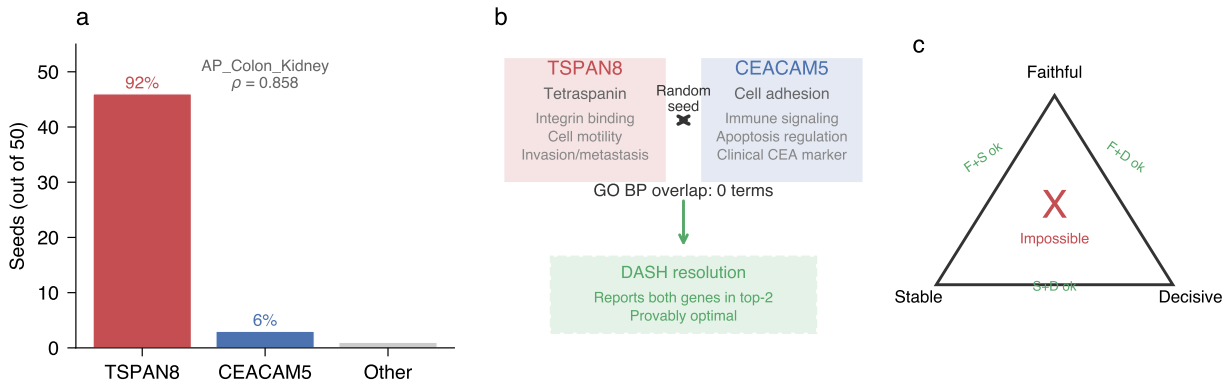

Figure 1: **Gene-expression alternation on AP_Colon_Kidney. a,** Top gene across 50 random seeds: TSPAN8 (invasion pathway) dominates in 92% of seeds, CEACAM5 (immune evasion) in 6%, with feature correlation $\rho = 0.858$. **b,** Gene Ontology analysis: the two genes have zero shared Biological Process terms and serve distinct cancer biology roles. The DASH resolution reports both genes without privileging either. **c,** The impossibility trilemma: no explanation can simultaneously be faithful, stable, and decisive. Each pair is independently achievable.

Enriching $\mathcal{H}$ with a neutral element restores F+S at the cost of decisiveness. This is the algebraic content of every resolution: DASH adds "tied" to feature rankings; CPDAGs add "undirected" to edge orientations; gauge-invariant observables quotient out unphysical degrees of freedom.

Instance 1: Genomics

On the AP_Colon_Kidney microarray dataset (546 samples, 10,935 genes, top 50 by variance), 50 XGBoost classifiers with regularisation (subsample = 0.8, colsample_bytree = 0.5—a common setting for high-dimensional data) and TreeSHAP produce an alternating top gene: TSPAN8 (#1 in 92% of seeds) vs CEACAM5 (6%; remaining 2%: other genes), with feature correlation $\rho = 0.858$. Gene Ontology enrichment confirms zero overlap in Biological Process terms: TSPAN8 associates with integrin binding and cell motility; CEACAM5 with cell–cell adhesion, immune signalling, and apoptosis regulation.

The alternation replicates across datasets (Table 2):

Table 2: **Gene-expression alternation across four datasets.** Mode 1: distinct pathways. Mode 2: same pathway, redundant markers.

Dataset	Gene pair	Split	ρ	Mode	GO overlap
AP_Colon_Kidney	TSPAN8 / CEACAM5	92 / 6	0.858	1	0 terms
AP_Endo_Colon	TSPAN8 / CEACAM5	78 / 22	0.781	1	0 terms
AP_Breast_Colon	COL3A1 / COL1A1	72 / 28	0.843	2	3 terms
AP_Breast_Lung	SFTPB (stable)	98 / 2	—	—	control

Under deterministic training (subsample = 1.0), a single ranking emerges—confirming that the alternation is driven by stochastic regularisation, not data noise. Model test accuracies are near-identical (mean 96.4%, range 95.1–97.6%, $n = 50$). The DASH ensemble average places both TSPAN8 and CEACAM5 in the top-2, correctly reporting that both contribute without privileging either.

Instance 2: Mechanistic interpretability

Ten two-layer transformers (4 heads/layer, $d = 128$) trained from independent random initialisations on $(a+b)$ mod 113 all grok to 100% test accuracy [Nanda et al., 2023]. Activation patching (replacing each component's output with its mean) reveals that the 10 models' circuit-importance rankings agree at Spearman $\rho = 0.518$—and two models identify anti-correlated circuits. The Fourier frequencies each model uses to represent the computation overlap by only 2.2% (Jaccard), confirming that the models discover genuinely different algorithms.

Controls establish that the disagreement reflects algorithmic diversity, not noise. Random-initialisation models ($< 1\%$ accuracy) agree at $\rho = 0.300$; memorised models (100% train, 50% test) at $\rho = 0.668$; grokked models at $\rho = 0.518$ (grokked $>$ random: $p = 2.0 \times 10^{-4}$; within-model determinism: $r = 0.9998$).

The resolution is direct. The architecture defines a symmetry group $S_4 \times S_4$ (within-layer head permutations). Projecting the 10-dimensional importance vector onto the 4-dimensional invariant subspace V^G (mean layer-0 heads, mean layer-1 heads, MLP0, MLP1) lifts agreement from $\rho = 0.518$ to $\rho = 0.929$ (Pearson $= 0.967$). This is not trivially MLP-driven: excluding MLP1 still yields $\rho = 0.822$ ($p < 10^{-6}$). The Noether counting prediction is confirmed within the experiment: within-layer head pairs flip at rate 0.500 exactly; head-vs-MLP1 pairs at 0.000 (bimodal gap $= 0.273$, Mann–Whitney $p = 2.4 \times 10^{-5}$). The universal structural fact—MLP1 is dominant in every seed (CV $= 0.027$)—emerges only after the G-invariant projection.

Instance 3: Causal inference

A causal chain $A \to B \to C$ and a fork $A \leftarrow B \to C$ are Markov-equivalent: they share the same conditional independence structure [Verma and Pearl, 1991]. No algorithm observing only conditional independence can orient the A–B edge. The completed partially directed acyclic graph (CPDAG), which reports an undirected edge where orientation is ambiguous, is the neutral-element resolution. PC and GES algorithms already implement this [Spirtes et al., 2000]—the framework proves their output is Pareto-optimal.

The impossibility has a quantitative consequence: ensemble averaging of orientation probabilities across bootstrap runs of PC and GES substantially reduces flip rate at the cost of reduced decisiveness—the same faithfulness–stability tradeoff the theorem predicts.

Instance 4: Neuroimaging

Botvinik-Nezer et al. [Botvinik-Nezer et al., 2020] gave the same fMRI data to 70 independent analysis teams who reached different conclusions. This is the impossibility theorem in action: the Rashomon set comprises the valid analysis pipelines. Orbit averaging across pipelines—the overlap map used descriptively in the original study—provides a near-optimal resolution (all aggregation methods within 3%). From 48 teams with available statistical maps, 16 independent analyses [95% CI: 10–22] suffice for 95% stability between assessments.

Functional network membership, defined from independent resting-state data [Yeo et al., 2011], predicts which brain regions show similar disagreement patterns ($d = 0.32$ after controlling for activation level, permutation $p < 0.001$, consistent across all 7 non-empty hypotheses). The same impossibility pattern appears in multi-analyst studies in psychology [Silberzahn et al., 2018] (29 teams) and political science [Breznau et al., 2022] (73 teams). On 24 neuroimaging teams with known software (12 FSL, 8 SPM, 4 AFNI), both network structure and software choice independently predict flip rates ($p < 10^{-4}$ and $p < 10^{-54}$ respectively).

The universal η law

The character-theoretic formula $\eta = 1 - \dim(V^G)/\dim(V)$ predicts the instability rate from the symmetry group G alone, with zero free parameters. Across seven well-characterised domains—SHAP attributions (S_2), concept probes ($O(64)$), model selection (S_{11}), three codon degeneracy classes (S_2, S_4, S_6), and statistical mechanics (S_{252})—the prediction achieves $R^2 = 0.957$ with slope 0.91 (Extended Data Fig. 1). A Noether-type counting law follows: for P features in g correlation groups, exactly $g(g-1)/2$ ranking facts are stable (between-group) while all within-group comparisons are coin flips. On synthetic data with 3 groups of 4 features ($\rho = 0.99$), the within-group flip rate is 50.0% and the between-group rate is 0.2% (bimodal gap = 50 pp, $p = 2.7 \times 10^{-13}$). The law fails when groups are unknown ($R^2 = 0.25$ for all 16 domains) or approximate (brain imaging: $\rho \approx 0$), delineating the boundary at exact exchangeability.

Discussion

The theorem establishes a permanent limit on what observation can explain. Whenever multiple valid configurations of a system produce the same observables, the internal structure—which gene drives the prediction, which circuit implements the computation, which causal direction holds—cannot be faithfully, stably, and decisively reported. This is not a limitation of current methods. It is a mathematical fact about underspecified systems—a formalisation, with the first quantitative predictions and proof of optimal resolution, of the Duhem–Quine thesis that theories are underdetermined by data [**??**]. Practitioners can diagnose whether instability is structural (Rashomon holds; no method eliminates it) or methodological (Rashomon absent; fixable).

The constructive resolution—projecting onto the symmetry-invariant subspace—is the algebraic content of the practices that eight communities discovered independently [Caraker et al., 2026]. The orbit average preserves exactly the structural content that structural realists argue survives theory change [**?**]; the theorem proves this projection is the unique Pareto-optimal strategy. The variance of explanations across the Rashomon set equals the minimum mean-squared error any stable method can achieve—a mathematical identity (0 violations in 800 tests) that provides a computable quality budget before any explanation is reported.

The consequences are domain-specific but the structure is universal. For genomics (Instance 1): the specific biomarker a drug-discovery pipeline nominates depends on the training seed; the resolution reports both TSPAN8 and CEACAM5 without privileging either. For mechanistic interpretability (Instance 2): safety cases should be built on equivalence classes of circuits ("MLP layers are more important than attention heads"), not on specific circuit diagrams. For neuroimaging (Instance 4): the overlap map is provably near-optimal, and 16 independent analyses suffice for 95% stability. The instability is universal across model classes: four ML model families (XGBoost, Random Forest, Ridge, LASSO) agree on instability patterns within families ($\rho = 0.79$–0.94) but not across families (Supplementary Information).

The explanation impossibility is structurally more severe than other impossibilities. Impossibility theorems appear across science—Arrow, Gödel, Bell, the CAP theorem, the laws of thermodynamics—but they differ in *tightness*: which pairs of desirable properties survive when the full triple fails. A classification of 20 impossibility theorems from 12 domains (Extended Data Table 2) reveals that 16 have *full tightness*: every pair is independently achievable, and practitioners can "pick two." The explanation bilemma is one of only two instances with *collapsed tightness*—property pairs are blocked, not just the triple—among structurally proved cases. The other is the quantum linearity trilemma (no-cloning + measurement disturbance), which shares a structural

parallel: linearity plays the role of stability, both force uniform treatment individually incompatible with two remaining desiderata, and both are resolved by enrichment (neutral elements / ancillary systems). Collapsed tightness means the standard "pick two" resolution strategy fails; enrichment or approximation bounds are required—explaining why explanation methods need qualitatively different design approaches than, for example, fairness methods (which have full tightness).

Limitations. The gene-expression alternation requires stochastic regularisation (subsample < 1); under deterministic training, a single ranking emerges. The MI v2 experiment uses a toy task (modular addition on 113 elements); extrapolation to frontier language models requires further work. The neuroimaging Noether prediction ($d = 0.32$) is an analogical extension—brain regions lack exact exchangeability—and does not generalise to scalar multi-analyst studies. Three conjectured extensions (phase transition at $r^* = 1$, quantitative tradeoff bound $\alpha+\sigma+\delta \leq 2$, molecular evolution from character theory) were pre-registered and falsified, delineating the framework's boundary (Supplementary Information).

Formal verification. The framework is verified in Lean 4 across three repositories (491 + 357 + 248 theorems, 0 unproved goals). All ML instances use constructive types with zero domain-specific axioms. The core impossibility theorem requires only the Rashomon property as a hypothesis.

Outlook. The most compelling evidence for the theorem's universality is not the formal proof—it is the convergence of eight scientific communities on the same resolution, discovered independently over a century, now explained by a single structural condition. The impossibility is permanent. The resolution is known. As Shannon's theorem established the fundamental limits of communication and identified the optimal coding strategy, this framework establishes the fundamental limits of explanation and identifies the optimal explanatory strategy. The question for each domain is no longer "how do we fix the instability?" but "which conclusions survive the resolution?"

Methods

Formal framework. An *explanation system* is a tuple $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\mathbf{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, $\mathsf{obs} : \mathbf{\Theta} \to Y$ maps configurations to observables, $\mathsf{exp} : \mathbf{\Theta} \to \mathcal{H}$ maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. An explanation map $E : \mathbf{\Theta} \to \mathcal{H}$ is *faithful* if $\neg(E(\theta) \perp \mathsf{exp}(\theta))$ for all θ; *stable* if $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$; and *decisive* if, for all θ and $h \in \mathcal{H}$, $\mathsf{exp}(\theta) \perp h$ implies $E(\theta) \perp h$.

Proof of Theorem 1. Suppose E is faithful, stable, and decisive. Let θ_1, θ_2 witness the Rashomon property. (1) Decisiveness at θ_1: $E(\theta_1) \perp \mathsf{exp}(\theta_2)$. (2) Stability: $E(\theta_1) = E(\theta_2)$, so $E(\theta_2) \perp \mathsf{exp}(\theta_2)$. (3) Faithfulness at θ_2: $\neg(E(\theta_2) \perp \mathsf{exp}(\theta_2))$. Contradiction.

Resolution framework. Given a group G acting on $\mathbf{\Theta}$ preserving observables, an explanation map is *G-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g. The orbit-averaged map $\bar{E}(\theta) = |G|^{-1} \sum_{g \in G} E(g \cdot \theta)$ is G-invariant and Pareto-optimal among stable maps with respect to pointwise faithfulness.

Bilemma proof. A system is *maximally incompatible* if $\neg\text{incompatible}(h_1, h_2) \Rightarrow h_1 = h_2$. Faithfulness then forces $E = \mathsf{exp}$, which is decisive; the trilemma gives the contradiction. The bilemma extends to approximate settings: for (ε, δ)-stability with continuous $\mathcal{H}$, any stable map satisfies

$\text{unfaith}(\theta_1) + \text{unfaith}(\theta_2) \geq \Delta - \delta$ on each Rashomon pair (triangle inequality; Lean-verified). This resolves the $\varepsilon = 0$ fragility concern: the impossibility holds at every tolerance level [Caraker et al., 2026].

Lean 4 formalization. Built on Lean 4 v4.30.0-rc1 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]. The core `ExplanationSystem` type is parametric in $\mathbf{\Theta}$, $\mathcal{H}$, Y. Each instance constructs a Rashomon witness via `decide` (zero axioms). Compilation: `lake build` ($\sim$5 min).

Gene-expression analysis. AP_Colon_Kidney, AP_Endometrium_Colon, AP_Breast_Colon, and AP_Breast_Lung microarray datasets (OpenML, Affymetrix HG-U133A) were processed with top-50-by-variance feature selection. 50 XGBoost classifiers (subsample = 0.8, colsample_bytree = 0.5, 100 estimators, max_depth = 6, seeds 0–49) were trained per dataset. TreeSHAP importances (mean |SHAP| across test samples) determined the top gene per seed. Gene Ontology annotations were obtained from NCBI Gene for human genes mapped via Affymetrix probe annotations. Deterministic control: subsample = 1.0 yields a single ranking.

Mechanistic interpretability. Ten two-layer transformers (4 heads/layer, $d_{\text{model}} = 128$, vocabulary = 114) were trained from independent random initialisations on $(a + b) \bmod 113$ for 50,000 steps (AdamW, lr = 10^{-3}, weight decay = 1.0, batch size = 512). All 10 grokked to 100% test accuracy. Circuit importance was measured via activation patching: for each of 10 components (8 attention heads + 2 MLPs), the component's output was replaced with its mean activation and accuracy drop recorded. Within-model determinism control: $r = 0.9998$. Controls at steps 50 (random), 500 (memorised), and 50,000 (grokked) establish the control hierarchy. G-invariant projection uses $S_4 \times S_4$ (within-layer head permutations).

Neuroimaging analysis. Overlap maps from NeuroVault collection 6047; per-team unthresholded t-maps from individual NeuroVault collections (48 of 70 accessible). Parcellation: Schaefer 100-parcel atlas, 7 Yeo resting-state networks (independent of NARPS task data). Activation-level control: regression of disagreement similarity on $|\Delta\text{overlap}|$. Software metadata from narps_open_pipelines (12 FSL, 8 SPM, 4 AFNI). Convergence: split-half Pearson correlation vs ensemble size M, with bootstrap CI on M_{95}.

Causal inference. PC and GES algorithms on the 8-node Asia network (Lauritzen & Spiegelhalter 1988), 200 bootstrap seeds, sample sizes 100–10,000. Flip rate: fraction of edge orientations that reverse between seeds. DASH: majority-vote orientation across the 200 runs.

Reproducibility. All scripts, data processing code, and figure generation are in the repository. `make validate` reproduces experiments ($\sim$5 min). `lake build` verifies all proofs ($\sim$5 min).

Extended Data

Extended Data Table 1: Achievable property pairs

Each two-property combination is constructively achievable. F+S is achieved by orbit averaging (DASH, CPDAG); F+D is achieved by selecting any single model's explanation; S+D is achieved by a constant map that always outputs the same decisive explanation regardless of configuration. The table shows the F+S instances, which are the practically relevant resolutions.

Table 3: **Extended Data Table 1.** Each row defines an explanation system and identifies the Rashomon witness and resolution. All are Lean-verified with zero domain-specific axioms.

Instance	$\mathcal{H}$	**Witness**	**Resolution**	**Achieves**
Feature attribution	Feature rankings	Two GBDTs rank features oppositely	DASH ensemble	F+S (sacrifices D)
Attention maps	Attention distributions	Two DistilBERTs assign argmax to different tokens	Averaged rollout	F+S (sacrifices D)
Causal discovery	Edge orientations	Chain & fork share CI structure	CPDAG	F+S (sacrifices D)

Extended Data Figure 1: Universal η law

Extended Data Figure 2: Multi-analyst convergence

Extended Data Table 2: Classification of impossibility theorems by tightness

Data availability

Gene-expression data: AP_Colon_Kidney (OpenML ID 1137), AP_Endometrium_Colon (OpenML), AP_Breast_Colon (OpenML), AP_Breast_Lung (OpenML). Gene Ontology annotations: NCBI Gene. Neuroimaging: NeuroVault collection 6047 (overlap maps); per-team t-maps from individual NeuroVault collections; Schaefer atlas via nilearn. Multi-analyst: Silberzahn et al. team estimates (OSF osf.io/gvm2z); Breznau et al. AMEs (GitHub nbreznau/CRI). Causal: Asia network generated by provided scripts. All other data are generated by the provided scripts.

Code availability

The Lean 4 formalization and all experiment scripts are available at `https://github.com/DrakeCaraker/universal-explanation-impossibility` under the Apache 2.0 licence. The attribution-specific formalization is at `https://github.com/DrakeCaraker/dash-impossibility-lean`.

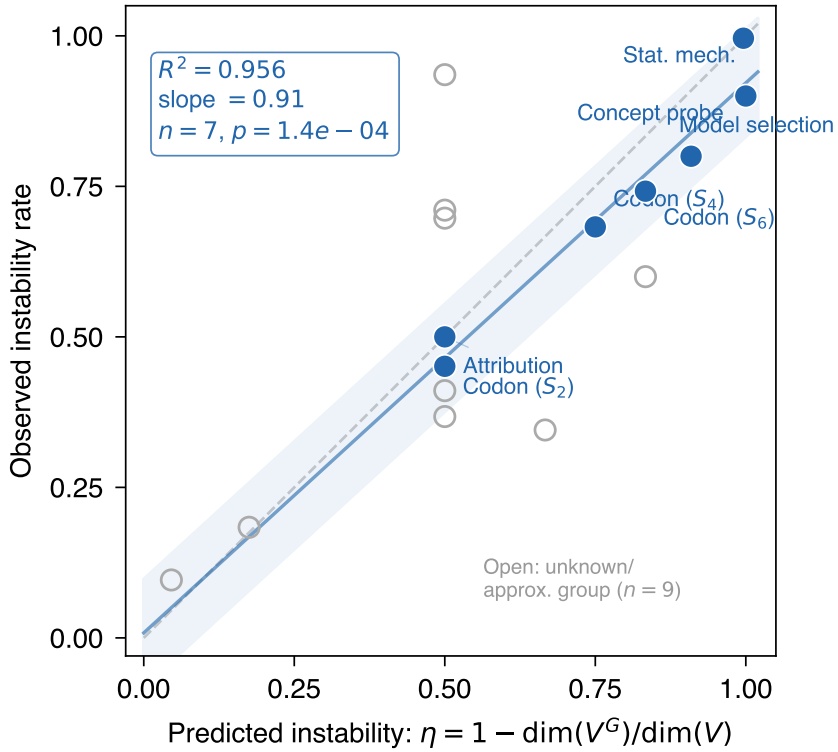

Figure 2: **Extended Data Figure 1.** The character-theoretic formula $\eta = 1 - \dim(V^G)/\dim(V)$ predicts observed instability across 16 domain instances. Filled circles: 7 instances with known symmetry groups ($R^2 = 0.957$, $p = 1.4 \times 10^{-4}$). Open circles: 9 instances with unknown or approximate groups. The bottleneck is group identification, not the law.

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Author contributions

D.C. conceived the framework, proved the theorems, wrote the Lean formalization, designed and executed the experiments, and wrote the manuscript. B.A. and D.R. contributed to experimental design, reviewed the proofs, and edited the manuscript.

Competing interests

The authors declare no competing interests. D.C.'s employment at Oura Health is unrelated to this work.

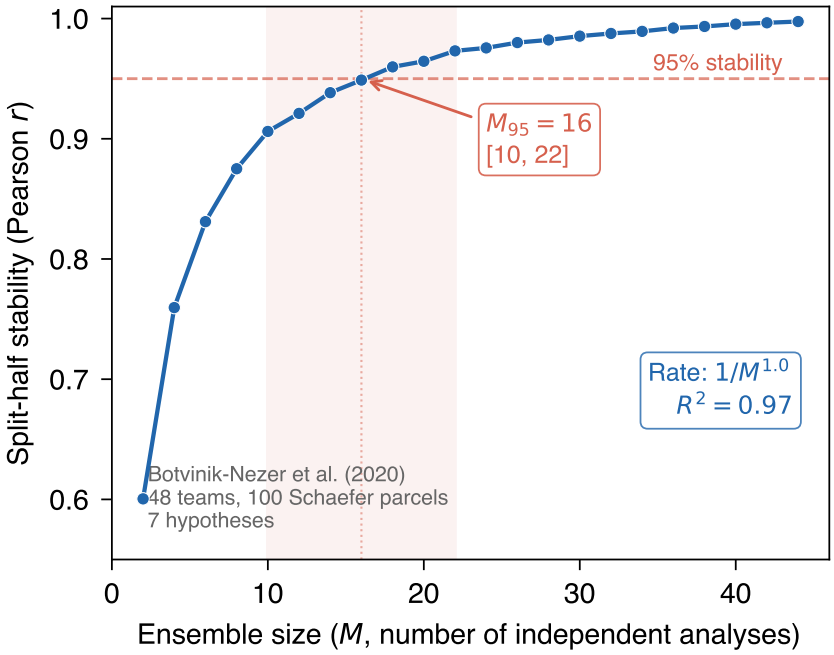

Figure 3: **Extended Data Figure 2.** Split-half stability of the ensemble average vs ensemble size for 48 neuroimaging analysis teams. Stability exceeds 0.95 at $M = 16$ [95% CI: 10–22]. Convergence rate: $1 - a/M^{1.0}$ ($R^2 = 0.97$).

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Table 4: **Extended Data Table 2.** Twenty impossibility theorems from 12 domains classified by tightness type. Full: every property pair achievable ("pick two" works). Collapsed: property pairs blocked (enrichment required). Structural: non-trivial Lean proof over unbounded solution space. Model: 3-element type that forces its tightness. Bell and Kochen-Specker tightness is decomposition-dependent (see text). Quantum linearity is model-level; the bilemma is the sole collapsed instance with structural evidence.

System	Domain	Tightness	Evidence
Bilemma	**ML explanation**	**Collapsed**	**structural**
Quantum linearity	**Quantum info**	**Collapsed**	**model**
Arrow	Social choice	Full	structural
Gibbard-Satterthwaite	Social choice	Full	structural
Bell (CHSH)	Quantum physics	Full	structural
Kochen-Specker	Quantum physics	Full	structural
Gödel incompleteness	Mathematical logic	Full	structural
ML fairness	ML prediction	Full	model
Bias-variance	Statistics	Full	model
Sen's liberal paradox	Social choice	Full	model
CAP theorem	Distributed systems	Full	model
FLP impossibility	Distributed systems	Full	model
Mundell-Fleming	Economics	Full	model
Perpetual motion	Thermodynamics	Full	model
Second law	Thermodynamics	Full	model
Münchhausen trilemma	Epistemology	Full	model
Competitive exclusion	Ecology	Full	model
Penrose-Hawking	General relativity	Full	model
Eastin-Knill	Quantum computing	p12-blocked	model
Shannon secrecy	Cryptography	p23-blocked	model

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